

Here are some key components and features of VST and ADAS:

1. **Collision avoidance systems.** Detects obstacles and provides warnings to prevent collisions. These include features like automatic emergency braking (AEB), forward collision warning, and lane departure warning.
2. **Adaptive cruise control (ACC).** Maintains a set speed but adjusts it based on the distance to the vehicle in front, ensuring a safe following distance. ACC enhances comfort during long drives by reducing the need for constant speed adjustments.
3. **Lane keeping assistance.** Monitors lane markings and provides steering input or alerts if the vehicle drifts out of its lane. This feature also enhances safety by preventing unintended lane departures.
4. **Blind spot detection.** Uses sensors to monitor the vehicle's blind spots and alerts the driver if there's a vehicle in the adjacent lane. This feature also aims to reduce the risk of collisions during lane changes.
5. **Parking assistance.** Utilizes sensors and cameras to help with parking by providing visual and/or auditory guidance. This can include features like automatic parallel parking.
6. **Traffic sign recognition.** Uses cameras to identify and interpret traffic signs, displaying relevant information to the driver. This feature assists in maintaining awareness of speed limits and other road regulations.
7. **Driver monitoring systems.** Monitors the driver's behavior and alerts them if signs of drowsiness or distraction are detected. This feature enhances safety by promoting driver attentiveness.
8. **Cross traffic alert.** Warns drivers of approaching traffic from the side, such as when backing out of parking spaces. This feature reduces the risk of collisions in situations with limited visibility.